

Exact Conditional Kernels and Variance Decompositions for Finite First Passage

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Abstract

We construct all exact conditional kernels $\Pr(T_j = b \mid T_i = a)$ for a frozen family of twenty first-passage variables on uniform permutations of sixteen named labels. Two-dimensional finite differences of the exact joint survival arrays give 400 nonnegative joint laws. Of 6800 candidate conditioning rows, 4980 have positive mass and 1820 are correctly left undefined. Every ordered pair satisfies cellwise Bayes balance and exact laws of total expectation and total variance; the twenty diagonal kernels are identities. The certificate is finite combinatorics under NO_BAD_EULER_OR_ROOT_NUMBER.

1 Joint inversion

Let $T_i \in \{0, \dots, 16\}$ be the first prefix time at which target H_i is hit. On the uniform space of $16!$ label permutations, write

$$S_{ij}(k, \ell) = \#\{T_i > k, T_j > \ell\}.$$

For threshold -1 , the corresponding boundary is the exact one-variable survival count, and $S_{ij}(-1, -1) = 16!$. Define

$$N_{ij}(a, b) = S_{ij}(a-1, b-1) - S_{ij}(a, b-1) - S_{ij}(a-1, b) + S_{ij}(a, b). \quad (1)$$

Lemma 1. *The array N_{ij} is the exact joint PMF in permutation counts. It is nonnegative, sums to $16!$, and its row and column marginals equal the frozen single-target laws of T_i and T_j .*

Proof. Equation (1) is two-dimensional finite Mobius inversion on the threshold grid. Summing over one coordinate telescopes to the relevant one-dimensional boundary. Summing both coordinates gives $16!$. Exact enumeration certifies nonnegativity in all $400 \cdot 17^2 = 115600$ cells. $\square$

2 Conditional-kernel theorem

Put $m_i(a) = \sum_b N_{ij}(a, b)$, independent of j . When $m_i(a) > 0$, define

$$K_{ij}(a, b) = \frac{N_{ij}(a, b)}{m_i(a)}.$$

When $m_i(a) = 0$, there is no conditional probability space; the certificate stores the probability row and all conditional moments as `null`.

Theorem 1. *Across all 400 ordered pairs, exactly 4980 of 6800 candidate rows define normalized kernels; the remaining 1820 are empty. Every cell obeys*

$$\Pr(T_i = a)K_{ij}(a, b) = \Pr(T_j = b)K_{ji}(b, a) = \frac{N_{ij}(a, b)}{16!}. \quad (2)$$

For every ordered pair,

$$\mathbb{E}[\mathbb{E}(T_j \mid T_i)] = \mathbb{E}(T_j), \quad (3)$$

$$\mathbb{E}[\text{Var}(T_j \mid T_i)] + \text{Var}(\mathbb{E}[T_j \mid T_i]) = \text{Var}(T_j). \quad (4)$$

Moreover $K_{ii}(a, b)$ is the identity kernel on every attainable time.

Proof. Positive row mass gives normalization directly. Joint transpose $N_{ij}(a, b) = N_{ji}(b, a)$ proves (2). Weighting each conditional mean by $m_i(a)/16!$ and interchanging finite sums proves total expectation. Expanding the conditional second moments and subtracting the squared mean proves total variance. For $i = j$, the two passage times coincide pointwise, so joint mass lies only on $a = b$. $\square$

certified object	count
ordered kernels / joint PMFs	400
joint cells and Bayes balances	115600
attainable / empty rows	4980 / 1820
total-expectation / total-variance identities	400 / 400
diagonal identity kernels	20

3 Independent certificate and scope

The checker reconstructs the complete canonical receipt independently. It also intersects packed support indicators to rebuild all 6800 synchronous survival cells $S_{ij}(k, k)$ directly from the single-target source, followed by factorial completion. SymPy verifies every defined rational row, moment, variance split, and reverse cell. Clean replay preserves

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and all 16 hostile mutations are rejected.

No arithmetic/local data, Euler factors, root numbers, automorphy, full Burnside ring or table of marks, or Hilbert–Polya operator is claimed.